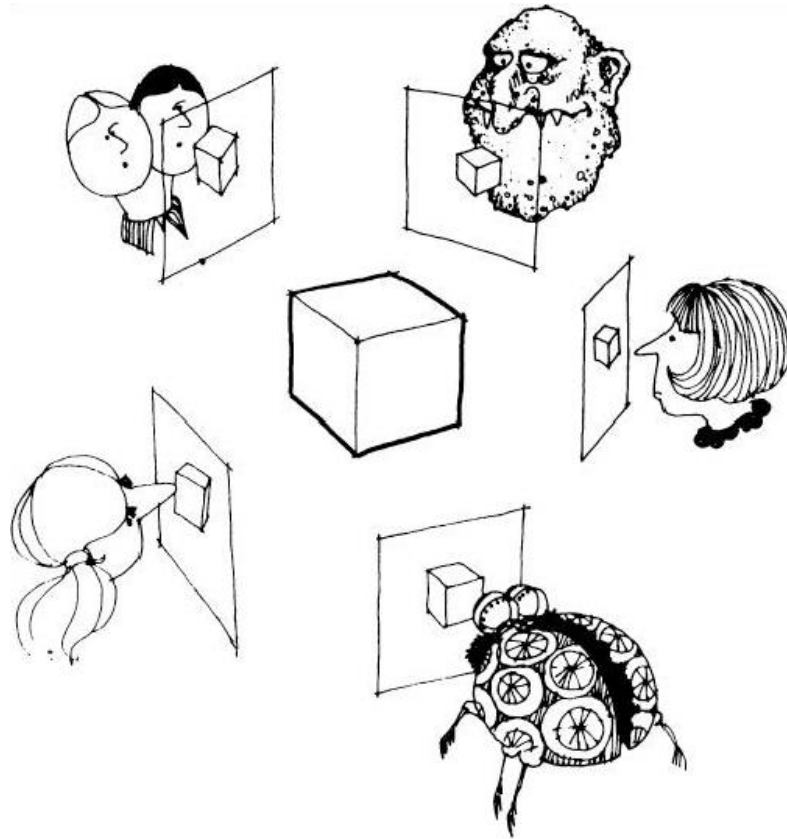


Perspective and 3D Geometry



Computer Vision

Adduru U G Sankararao, IIIT Sri City

Perspective and 3D Geometry

- **Epipolar Geometry and Stereo Vision**

- What's the mapping from two images taken with camera translation?

- **Structure from motion**

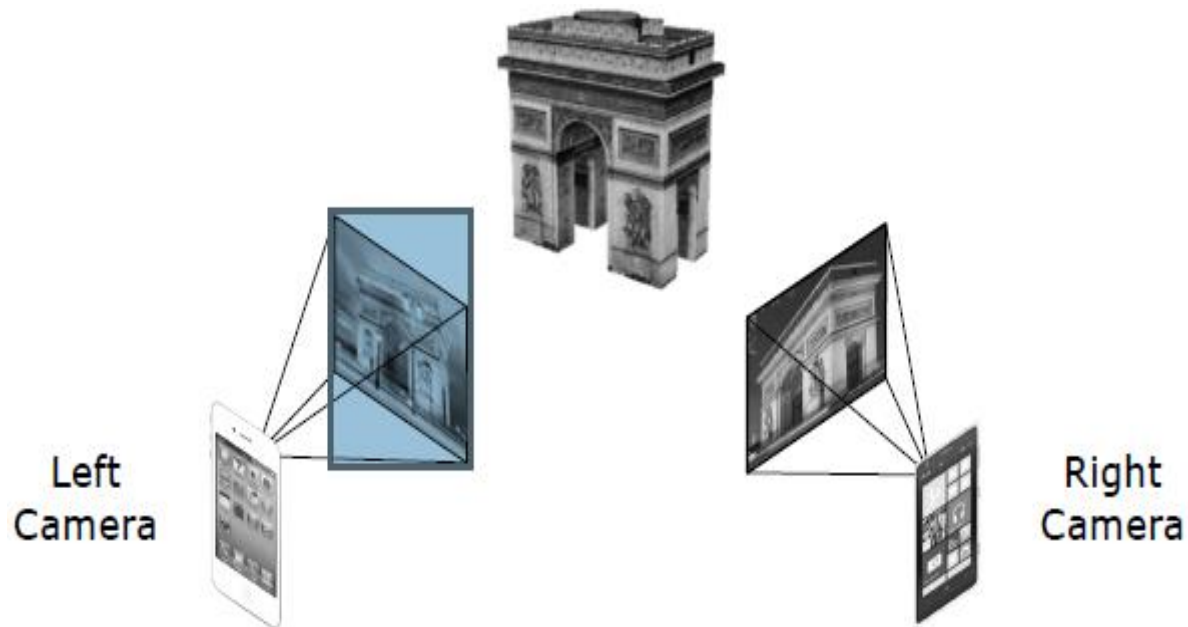
- How can we recover 3D points from multiple images?

Uncalibrated Stereo

Uncalibrated Stereo

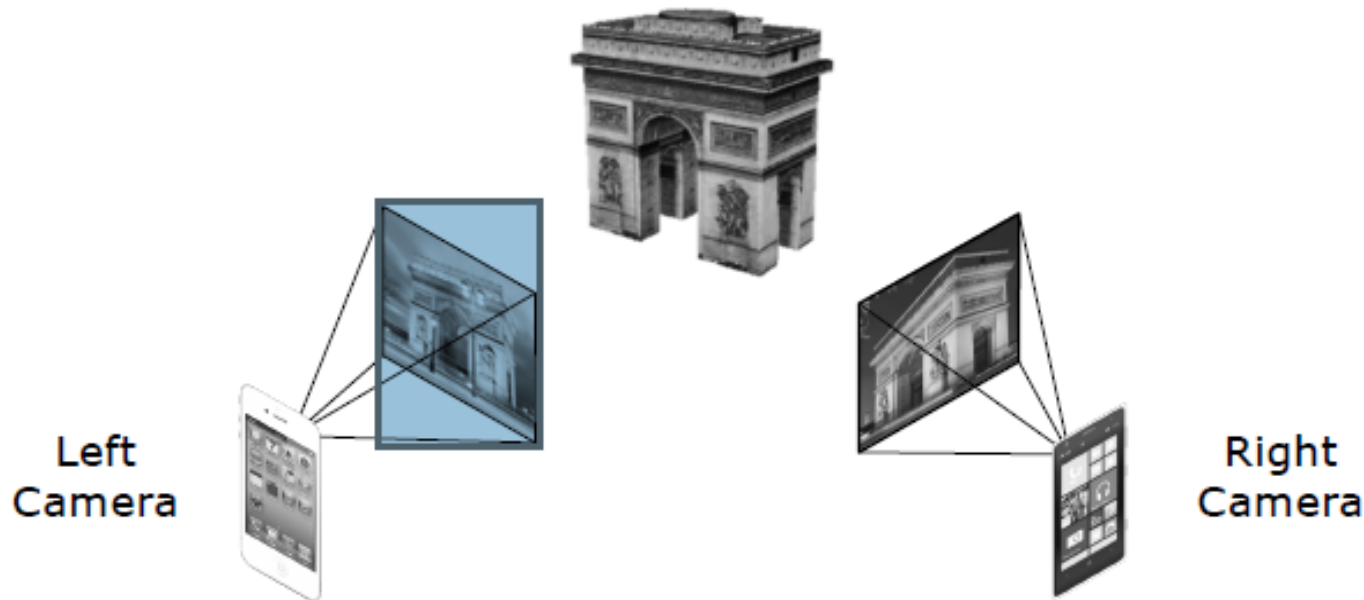
Method to estimate 3D structure of a static scene from two arbitrary views.

- Problem of Uncalibrated Stereo
- Epipolar Geometry
- Estimating Fundamental Matrix
- Finding Dense Correspondences
- Computing Depth



Uncalibrated Stereo

Compute 3D structure of static scene from two arbitrary views

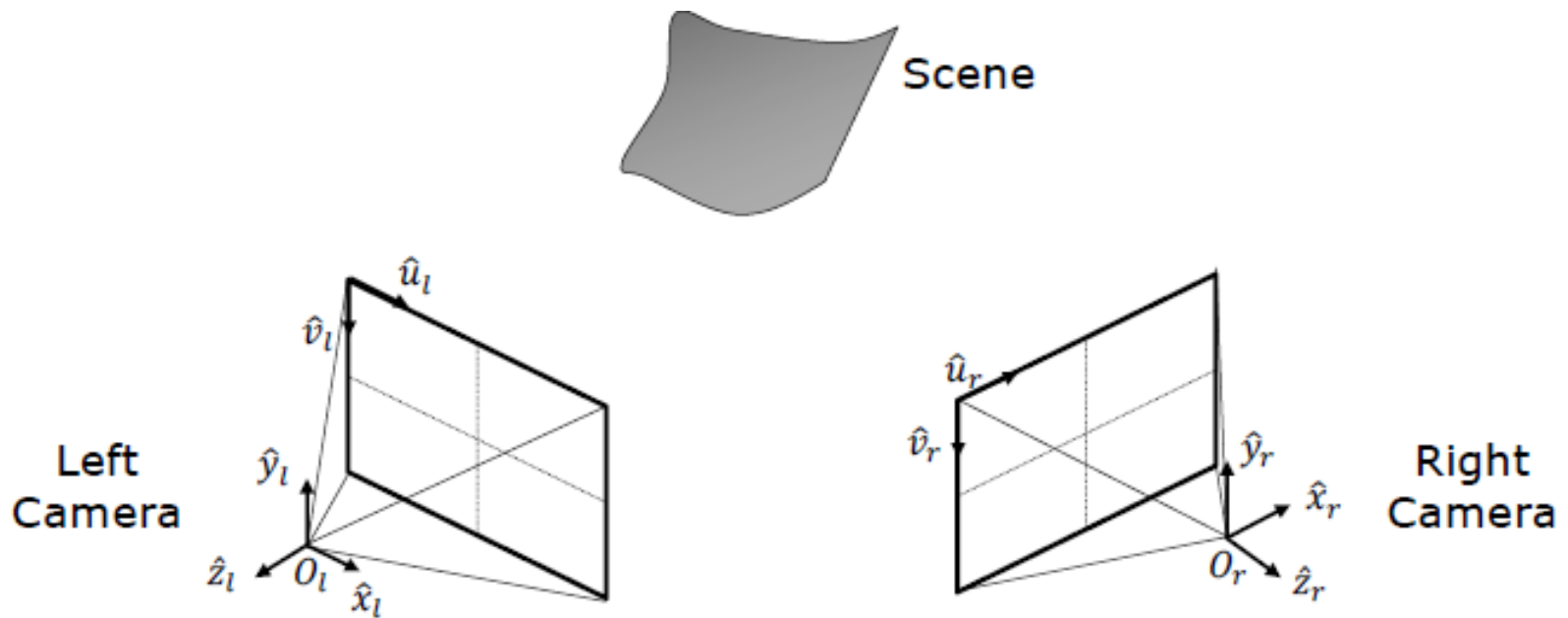


Intrinsics (f_x, f_y, o_x, o_y) are known for both views/cameras.

Extrinsics (relative position/orientation of cameras) are unknown.

- Calibrate the camera or retrieve the internal parameters from the meta-tag embedded in each image.
- Need to calibrate the uncalibrated stereo i.e., compute relative position and orientation of one camera w.r.t. other.

Uncalibrated Stereo



- ✓ 1. Assume Camera Matrix K is known for each camera
- 2. Find a few Reliable Corresponding Points

Initial Correspondence

Find a set of corresponding features (at least 8) in left and right images (e.g. using SIFT or hand-picked).

Left image



$$\bullet (u_l^{(1)}, v_l^{(1)})$$

$$\vdots$$

$$\bullet (u_l^{(m)}, v_l^{(m)})$$

Right image

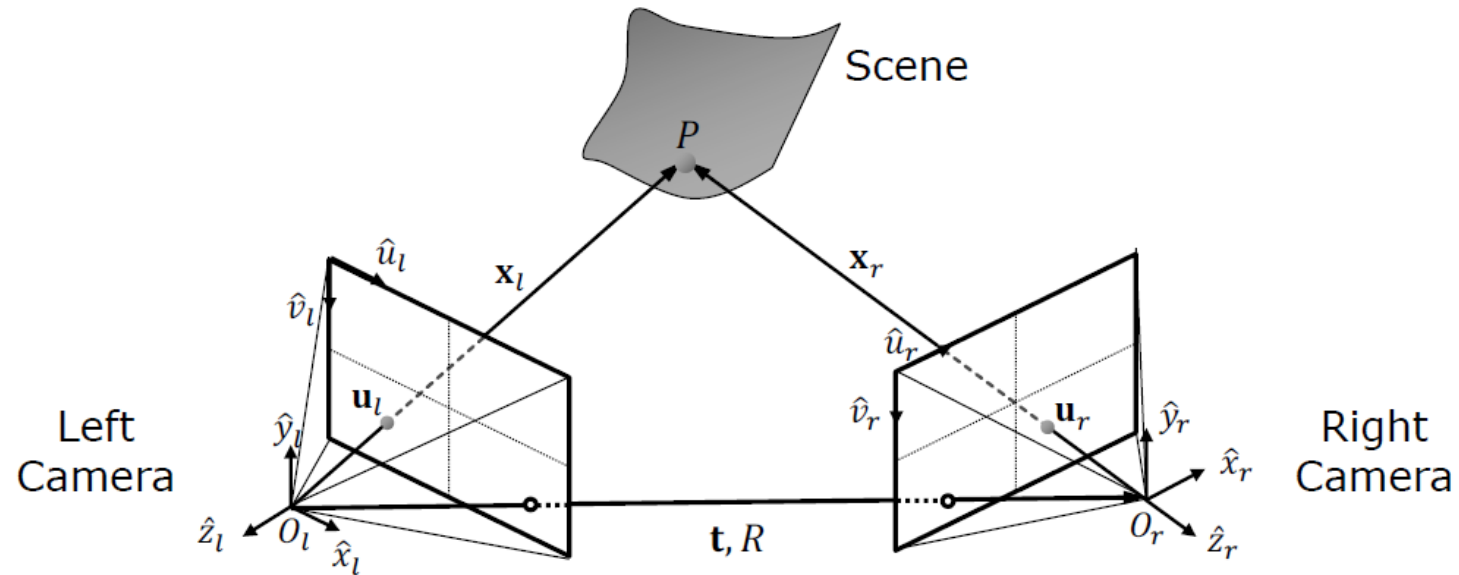


$$\bullet (u_r^{(1)}, v_r^{(1)})$$

$$\vdots$$

$$\bullet (u_r^{(m)}, v_r^{(m)})$$

Uncalibrated Stereo

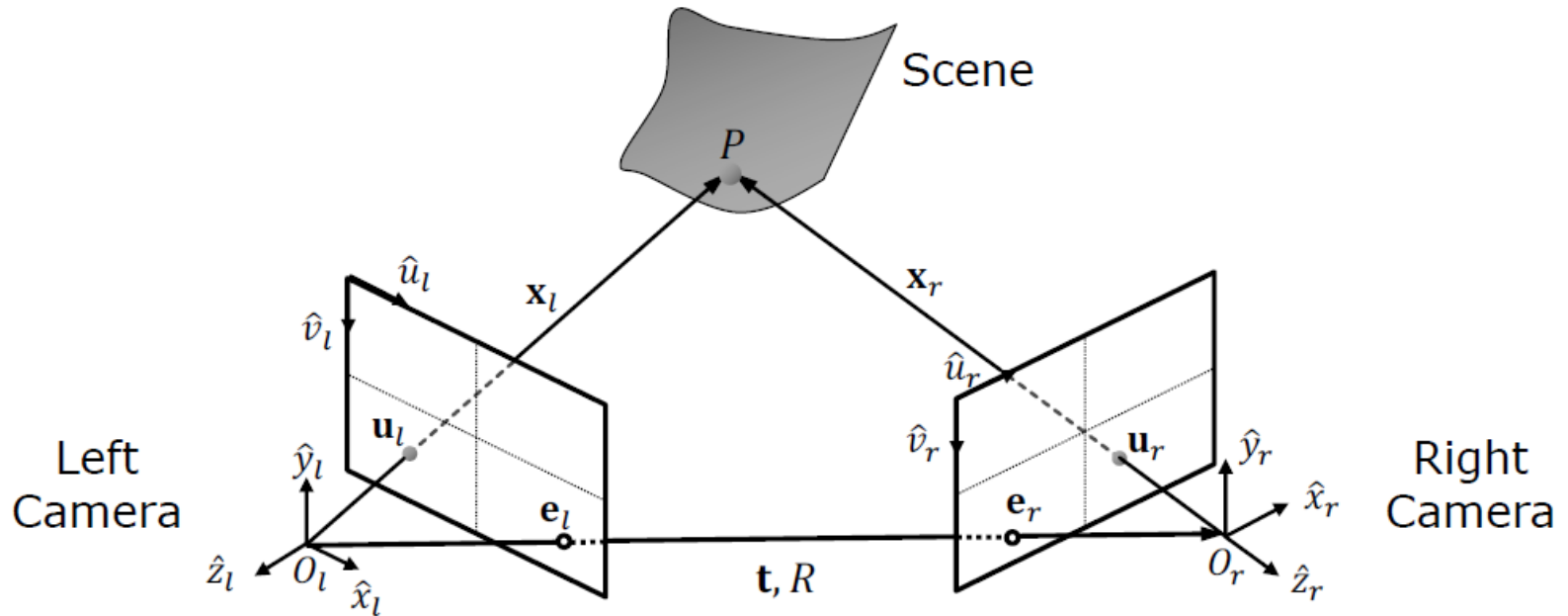


1. Assume Camera Matrix K is known for each camera
2. Find a few Reliable Corresponding Points
3. Find Relative Camera Position t and Orientation R
4. Find Dense Correspondence
5. Compute Depth using Triangulation

Epipolar Geometry

- Our goal is to compute relative position and orientation of one camera w.r.t. other
- This can be computed using Epipolar Geometry

Epipolar Geometry: Epipoles



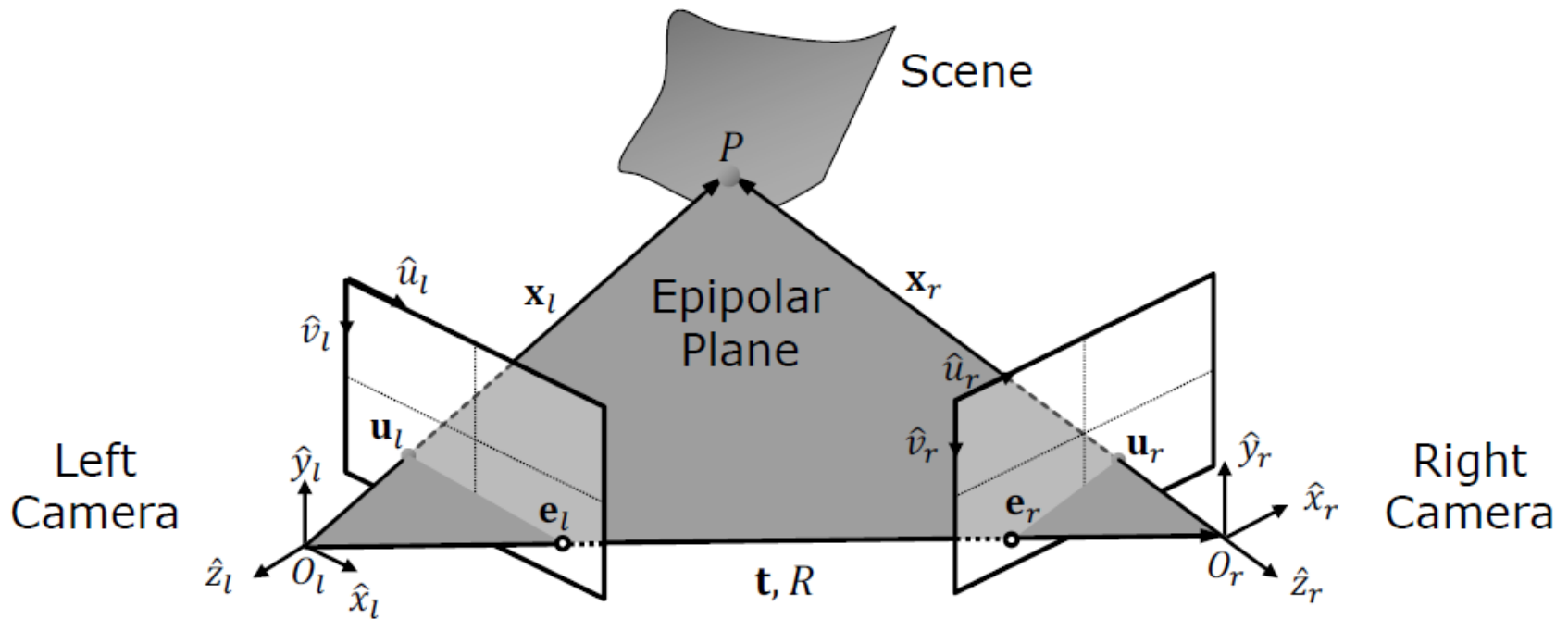
Epipole: Image point of origin/pinhole of one camera as viewed by the other camera.

e_l and e_r are the epipoles.

e_l and e_r are unique for a given stereo pair.

- In our figure, the epipoles happen to lie within the right and left images
- in general, the epipoles could lie outside the images as well.

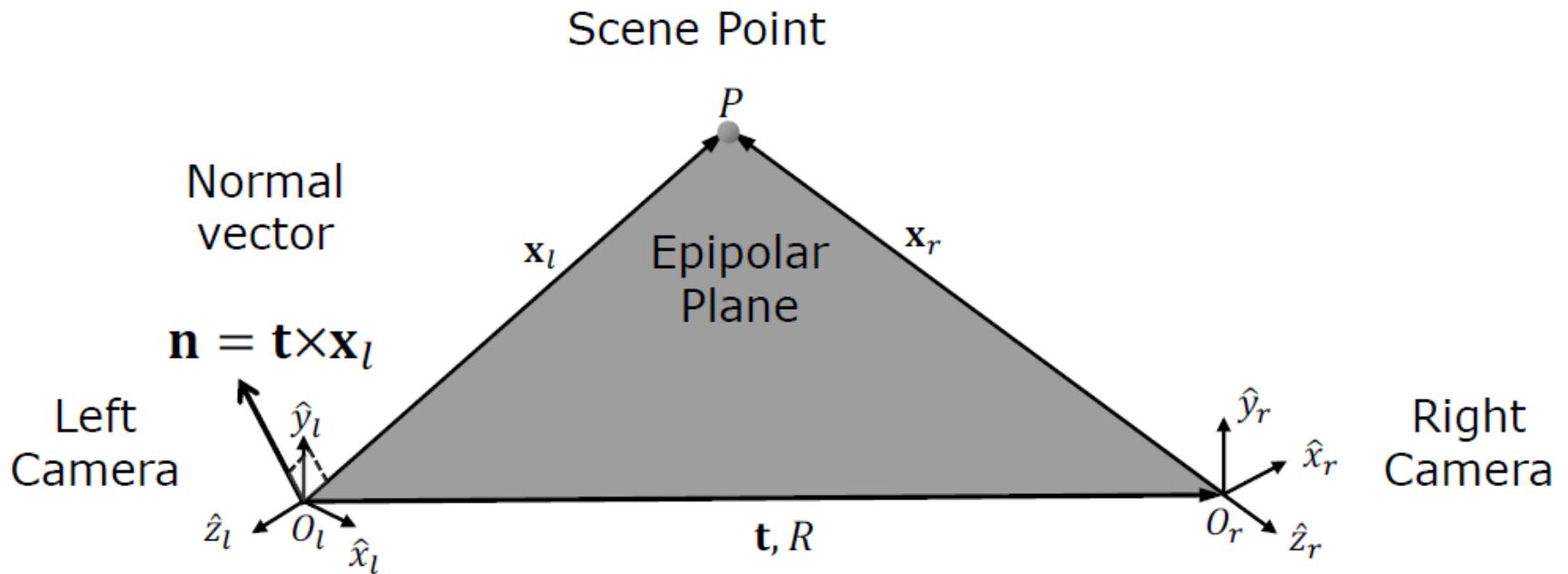
Epipolar Geometry: Epipolar Plane



Epipolar Plane of Scene Point P : The plane formed by camera origins (O_l and O_r), epipoles (e_l and e_r) and scene point P .

Every scene point lies on a unique epipolar plane.

Epipolar Constraint



Vector normal to the epipolar plane: $\mathbf{n} = \mathbf{t} \times \mathbf{x}_l$

Dot product of $\mathbf{n}$ and $\mathbf{x}_l$ (perpendicular vectors) is zero:

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

Epipolar Constraint

Writing the epipolar constraint in matrix form:

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

$$[x_l \quad y_l \quad z_l] \begin{bmatrix} t_y z_l - t_z y_l \\ t_z x_l - t_x z_l \\ t_x y_l - t_y x_l \end{bmatrix} = 0 \quad \text{Cross-product definition}$$

$$[x_l \quad y_l \quad z_l] \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = 0 \quad \boxed{1} \quad \text{Matrix-vector form}$$

$T_{\times}$

$\mathbf{t}_{3 \times 1}$: Position of Right Camera in Left Camera's Frame

$R_{3 \times 3}$: Orientation of Left Camera in Right Camera's Frame

$$\boxed{\mathbf{x}_l = R\mathbf{x}_r + \mathbf{t}} \quad \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \quad \boxed{2}$$

Epipolar Constraint

Substituting into the epipolar constraint gives:

$$[x_l \quad y_l \quad z_l] \left(\begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \right) = 0$$

3

$t \times t = 0$

$$[x_l \quad y_l \quad z_l] \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Essential Matrix E

$$E = T_{\times} R$$

- E relates $\mathbf{X}_l$ and $\mathbf{X}_r$

[Longuet-Higgins 1981]

Essential Matrix E: Decomposition

$$E = T_{\times} R$$

$$\begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

Given that $T_{\times}$ is a Skew-Symmetric matrix ($a_{ij} = -a_{ji}$) and R is an Orthonormal matrix, it is possible to “decouple” $T_{\times}$ and R from their product using “Singular Value Decomposition”.

Take Away: If E is known, we can calculate $\mathbf{t}$ and R .

How do we find E ?

Relates 3D position (x_l, y_l, z_l) of scene point w.r.t left camera to its 3D position (x_r, y_r, z_r) w.r.t. right camera

$$\boxed{\mathbf{x}_l^T E \mathbf{x}_r = 0}$$

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

3D position in left
camera coordinates

3x3 Essential
Matrix

3D position in right
camera coordinates

Unfortunately, we don't have $\mathbf{x}_l$ and $\mathbf{x}_r$.

But we do know corresponding points in image coordinates.

So, our goal is to recast our epipolar constraint in terms of (u_l, v_l) and (u_r, v_r) .

Incorporating the Image Coordinates

Perspective projection equations for left camera:

$$u_l = f_x^{(l)} \frac{x_l}{z_l} + o_x^{(l)} \qquad v_l = f_y^{(l)} \frac{y_l}{z_l} + o_y^{(l)}$$

$$z_l u_l = f_x^{(l)} x_l + z_l o_x^{(l)} \qquad z_l v_l = f_y^{(l)} y_l + z_l o_y^{(l)}$$

Representing in matrix form: Using homogenous coordinates

$$\boxed{1} \quad z_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = \begin{bmatrix} z_l u_l \\ z_l v_l \\ z_l \end{bmatrix} = \begin{bmatrix} f_x^{(l)} x_l + z_l o_x^{(l)} \\ f_y^{(l)} y_l + z_l o_y^{(l)} \\ z_l \end{bmatrix} = \underbrace{\begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} \\ 0 & f_y^{(l)} & o_y^{(l)} \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Known Camera Matrix } K_l} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix}$$

- Similarly for right camera also

Incorporating the Image Coordinates

Left camera

$$z_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} \\ 0 & f_y^{(l)} & o_y^{(l)} \\ 0 & 0 & 1 \end{bmatrix}}_{K_l} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix}$$

2

Right camera

$$z_r \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} \\ 0 & f_y^{(r)} & o_y^{(r)} \\ 0 & 0 & 1 \end{bmatrix}}_{K_r} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}$$

$$\mathbf{x}_l^T = [u_l \quad v_l \quad 1] z_l K_l^{-1^T}$$

$$\mathbf{x}_r = K_r^{-1} z_r \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}$$

Incorporating the Image Coordinates

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \cancel{K_l^{-1}^T} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} K_r^{-1} \cancel{z_r} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0 \quad \boxed{1}$$

The scene point is always away from the pinhole

$$z_l \neq 0$$

$$z_r \neq 0$$

- Here, elements of E and z_l, z_r are unknown

Incorporating the Image Coordinates

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} K_l^{-1T} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} K_r^{-1} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

2

Fundamental matrix F

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F

$$E = K_l^T F K_r$$

$$E = T_{\times} R$$

[Fagueras 1992, Luong 1992]

- If we can find F , we can easily compute the essential matrix E since K_l and K_r are known.
- From E , we can find the translation T_x and rotation R using singular value decomposition.

Estimating Fundamental matrix (F)

Stereo Calibration Procedure

Find a set of corresponding features in left and right images (e.g. using SIFT or hand-picked)

Left image



$$\bullet (u_l^{(1)}, v_l^{(1)})$$

$$\vdots$$

$$\bullet (u_l^{(m)}, v_l^{(m)})$$

Right image



$$\bullet (u_r^{(1)}, v_r^{(1)})$$

$$\vdots$$

$$\bullet (u_r^{(m)}, v_r^{(m)})$$

Stereo Calibration Procedure

Step A: For each correspondence i , write out epipolar constraint.

$$\underbrace{\begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix}}_{\text{Known}} \underbrace{\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} u_r^{(i)} \\ v_r^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}} = 0$$

Expand the matrix to get linear equation:

$$(f_{11}u_r^{(i)} + f_{12}v_r^{(i)} + f_{13})u_l^{(i)} + (f_{21}u_r^{(i)} + f_{22}v_r^{(i)} + f_{23})v_l^{(i)} + f_{31}u_r^{(i)} + f_{32}v_r^{(i)} + f_{33} = 0$$

Stereo Calibration Procedure

Step B: Rearrange terms to form a linear system.

$$\begin{array}{c}
 \left[\begin{array}{cccccccc|c}
 u_l^{(1)} u_r^{(1)} & u_l^{(1)} v_r^{(1)} & u_l^{(1)} & v_l^{(1)} u_r^{(1)} & v_l^{(1)} v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 u_l^{(i)} u_r^{(i)} & u_l^{(i)} v_r^{(i)} & u_l^{(i)} & v_l^{(i)} u_r^{(i)} & v_l^{(i)} v_r^{(i)} & v_l^{(i)} & u_l^{(i)} & u_r^{(i)} & 1 \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 u_l^{(m)} u_r^{(m)} & u_l^{(m)} v_r^{(m)} & u_l^{(m)} & v_l^{(m)} u_r^{(m)} & v_l^{(m)} v_r^{(m)} & v_l^{(m)} & u_l^{(m)} & u_r^{(m)} & 1
 \end{array} \right]
 \begin{array}{c}
 \left[\begin{array}{c}
 f_{11} \\
 f_{21} \\
 f_{31} \\
 f_{21} \\
 f_{22} \\
 f_{23} \\
 f_{31} \\
 f_{32} \\
 f_{33}
 \end{array} \right]
 \end{array}
 =
 \begin{array}{c}
 \left[\begin{array}{c}
 0 \\
 \vdots \\
 0 \\
 \vdots \\
 0
 \end{array} \right]
 \end{array}
 \\
 \text{\textbf{A}} \qquad \qquad \qquad \text{\textbf{f}} \\
 \text{(Known)} \qquad \qquad \qquad \text{(Unknown)}
 \end{array}$$

$$\boxed{\text{\textbf{A}} \text{\textbf{f}} = \text{\textbf{0}}}$$

Scale Issue

Fundamental matrix acts on homogenous coordinates.

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0 = \begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} kf_{11} & kf_{12} & kf_{13} \\ kf_{21} & kf_{22} & kf_{23} \\ kf_{31} & kf_{32} & kf_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}$$

Fundamental Matrix F and kF describe the same epipolar geometry. That is, F is defined only up to a scale.

Set Fundamental Matrix to some arbitrary scale.

$$\|\mathbf{f}\|^2 = 1$$

Solving for F

Step C: Find least squares solution for fundamental matrix F .

We want $A\mathbf{f}$ as close to 0 as possible and $\|\mathbf{f}\|^2 = 1$:

$$\min_{\mathbf{f}} \|A\mathbf{f}\|^2 \quad \text{such that } \|\mathbf{f}\|^2 = 1$$

Constrained linear least squares problem

Like solving Projection Matrix during Camera Calibration.

- Rearrange solution $\mathbf{f}$ to form the fundamental matrix F

Extracting Rotation and Translation

Step D: Compute essential matrix E from known left and right intrinsic camera matrices and fundamental matrix F .

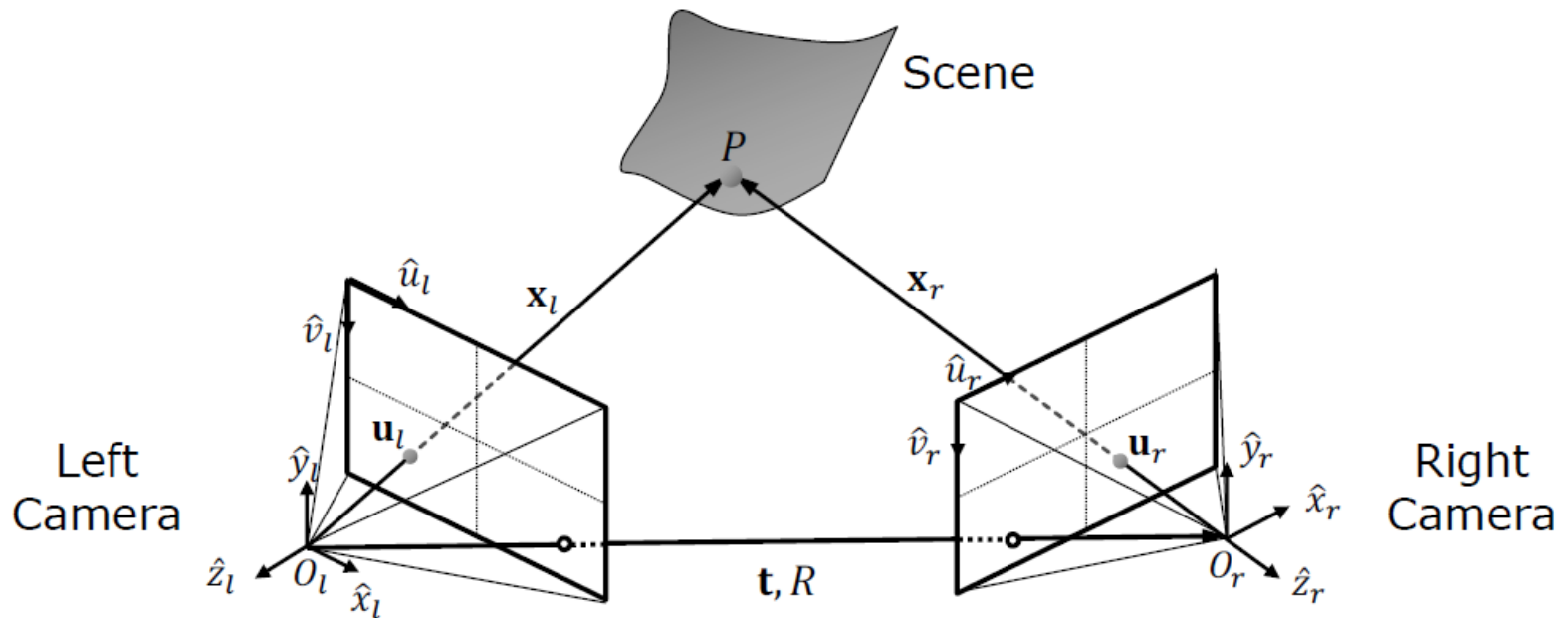
$$E = K_l^T F K_r$$

Step E: Extract R and t from E .

$$E = T_{\times} R$$

(Using Singular Value Decomposition)

Uncalibrated Stereo



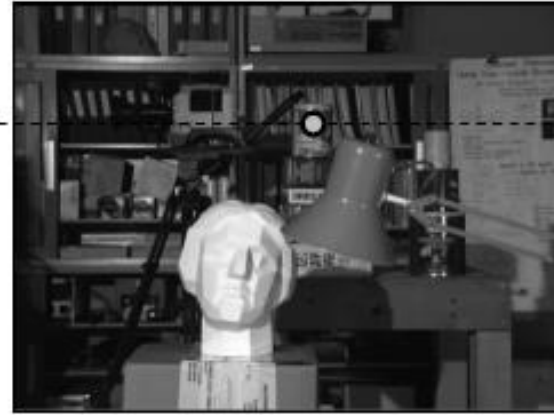
- ✓ 1. Assume Camera Matrix K is known for each camera
- ✓ 2. Find a few Reliable Corresponding Points
- ✓ 3. Find Relative Camera Position $\mathbf{t}$ and Orientation R
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- 5. Compute Depth using Triangulation

Finding Correspondences

Simple Stereo: Finding Correspondences (Recap)



Left Camera Image

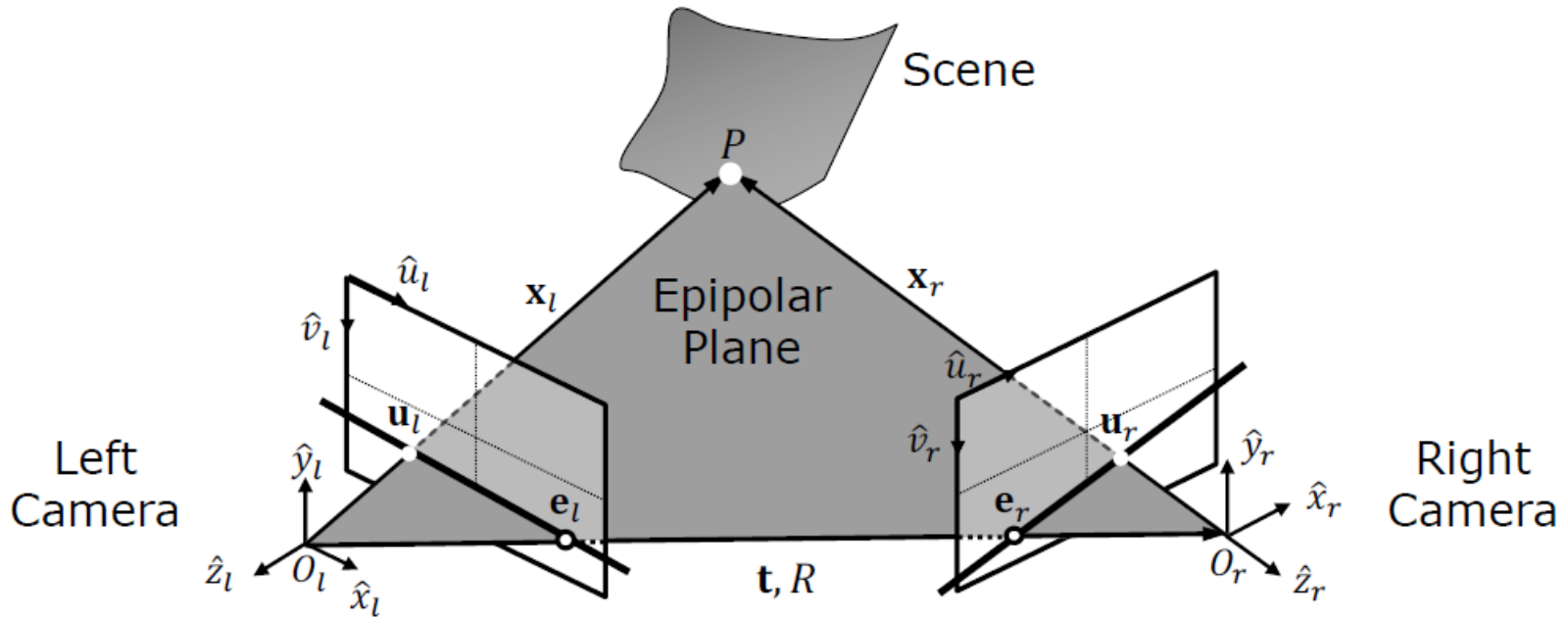


Right Camera Image

Corresponding scene points lie on the same horizontal scan-line.
Finding correspondence is a 1D search.

- If we can find Translation and Rotation matrices, then it turns out to be a 1D search.
- But along which line/direction, we need to search ?

Epipolar Geometry: Epipolar Line



Epipolar Line: Intersection of image plane and epipolar plane.

Every scene point has two corresponding epipolar lines, one each on the two image planes.

Finding Epipolar Lines

Given: Fundamental matrix F and point on left image (u_l, v_l)

Find: Equation of Epipolar line in the right image

Epipolar Constraint Equation:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Expanding the matrix equation gives:

$$(f_{11}u_l + f_{21}v_l + f_{31})u_r + (f_{12}u_l + f_{22}v_l + f_{32})v_r + (f_{13}u_l + f_{23}v_l + f_{33}) = 0$$

Equation for right epipolar line: $a_l u_r + b_l v_r + c_l = 0$

Similarly we can calculate epipolar line in left image for a point in right image.

Finding Epipolar Lines: Example

Given the Fundamental matrix,

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

and the left image point

$$\tilde{\mathbf{u}}_l = \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix}$$

Left Image



Right Image



The equation for the epipolar line in the right image is

$$\begin{bmatrix} u_r & v_r & 1 \end{bmatrix} \begin{bmatrix} -.003 & -.003 & 2.97 \\ -.028 & -.008 & 56.38 \\ 13.19 & -29.2 & -9999 \end{bmatrix} \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix} = 0$$

Finding Epipolar Lines: Example

Given the Fundamental matrix,

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

and the left image point

$$\tilde{\mathbf{u}}_l = \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix}$$



The equation for the epipolar line in the right image is

$$.03u_r + .99v_r - 265 = 0$$

Finding Correspondence



Left Image

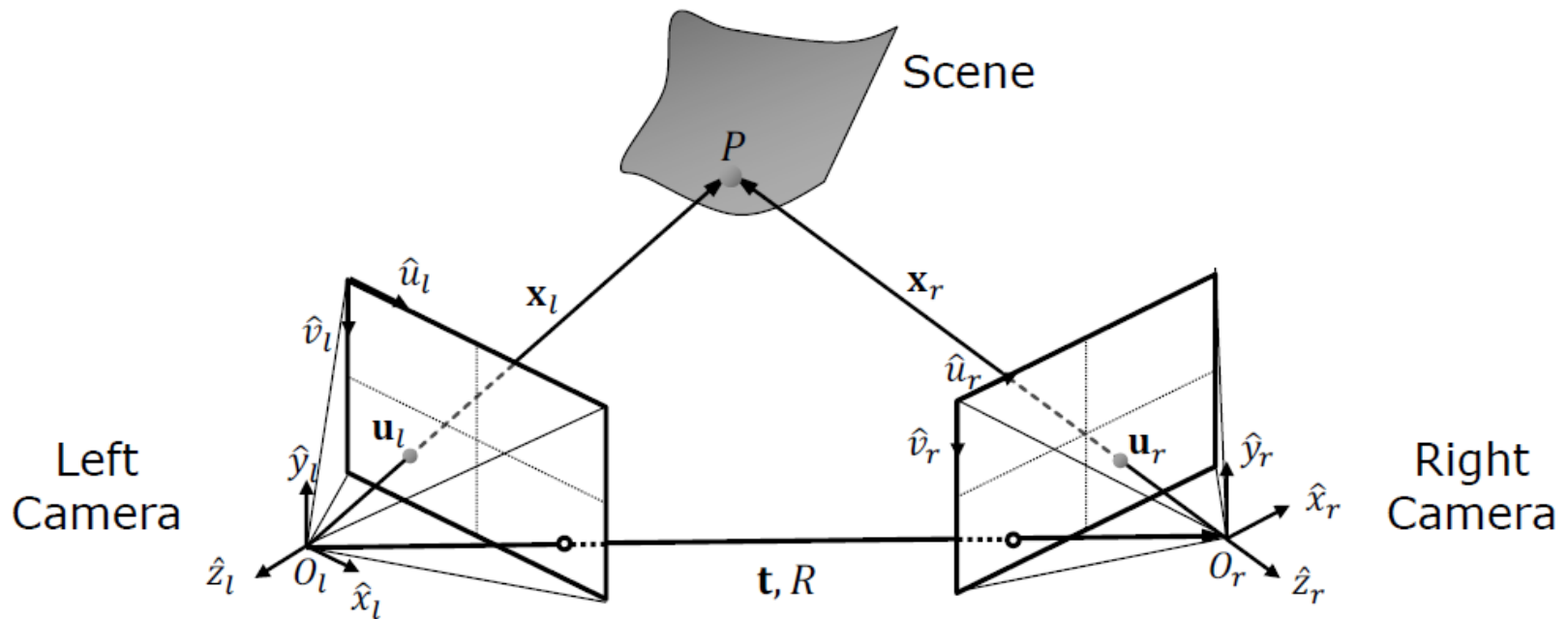


Epipolar
Line

Right Image

Corresponding scene points lie on the epipolar lines.
Finding correspondence is a 1D search.

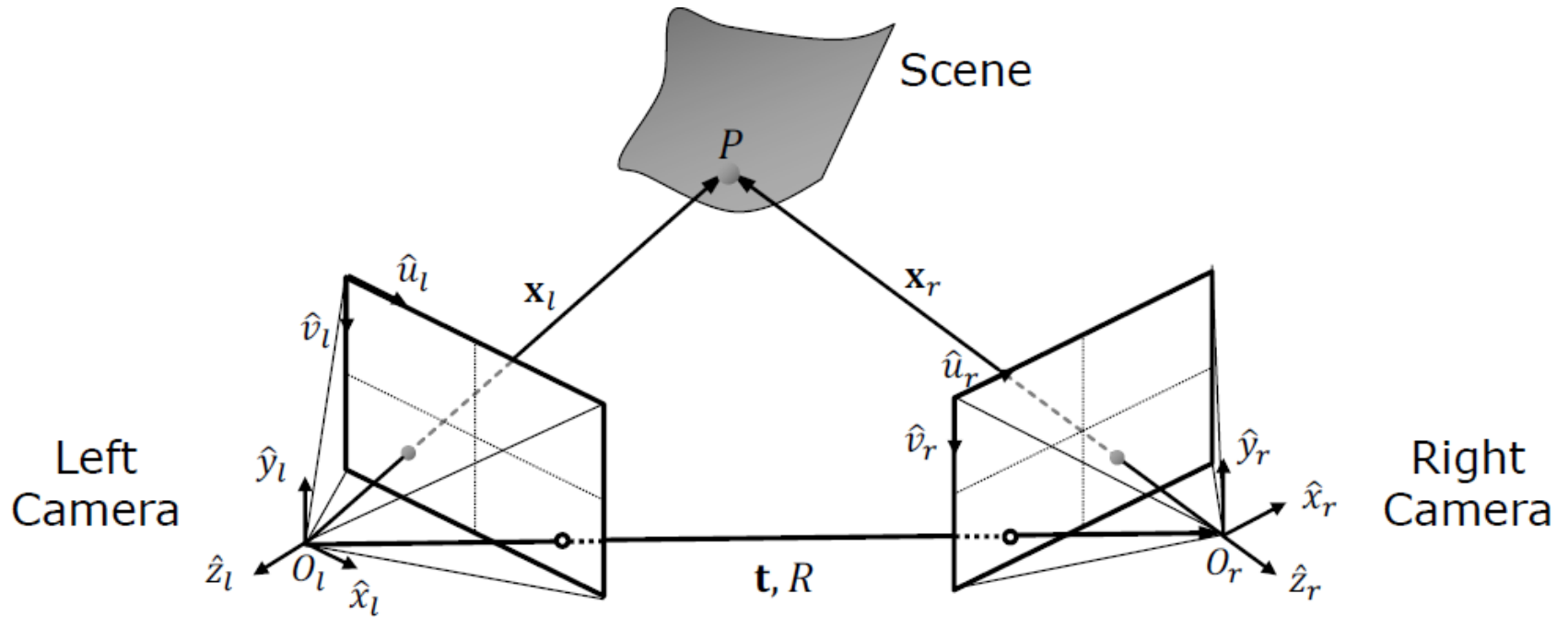
Uncalibrated Stereo



- ✓ 1. Assume Camera Matrix K is known for each camera
- ✓ 2. Find a few Reliable Corresponding Points
- ✓ 3. Find Relative Camera Position $\mathbf{t}$ and Orientation R
- ✓ 4. Find Dense Correspondence
- 5. Compute Depth using Triangulation

Computing Depth

Computing Depth



Given the intrinsic parameters, the projections of scene point on the two image sensors are:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Computing Depth

Left Camera Imaging Equation

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix}$$

Right Camera Imaging Equation

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

1

We also know the relative position and orientation between the two cameras.

$$\begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix} \quad 2$$

Computing Depth

Left Camera Imaging Equation:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix} \quad \boxed{3}$$

$$\tilde{\mathbf{u}}_l = P_l \tilde{\mathbf{x}}_r$$

P_l is the projection matrix
for left camera

Right Camera Imaging Equation:

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix} \quad \boxed{4}$$

$$\tilde{\mathbf{u}}_r = M_{int_r} \tilde{\mathbf{x}}_r$$

M_{int_r} is the intrinsic matrix
for right camera

Computing Depth

The imaging equations:

$$\tilde{\mathbf{u}}_r = M_r \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known

Unknown

$$\tilde{\mathbf{u}}_l = P_l \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known

Unknown

Rearranging the terms:

$$\begin{bmatrix} u_r m_{31} - m_{11} & u_r m_{32} - m_{12} & u_r m_{33} - m_{13} \\ v_r m_{31} - m_{21} & v_r m_{32} - m_{22} & v_r m_{33} - m_{23} \\ u_l p_{31} - p_{11} & u_l p_{32} - p_{12} & u_l p_{33} - p_{13} \\ v_l p_{31} - p_{21} & v_l p_{32} - p_{22} & v_l p_{33} - p_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} m_{14} - u_r m_{34} \\ m_{24} - v_r m_{34} \\ p_{14} - u_l p_{34} \\ p_{24} - v_l p_{34} \end{bmatrix} \boxed{1}$$

Computing Depth: Least Square Solution

$$\begin{array}{ccc}
 \boxed{\begin{bmatrix} u_r m_{31} - m_{11} & u_r m_{32} - m_{12} & u_r m_{33} - m_{13} \\ v_r m_{31} - m_{21} & v_r m_{32} - m_{22} & v_r m_{33} - m_{23} \\ u_l p_{31} - p_{11} & u_l p_{32} - p_{12} & u_l p_{33} - p_{13} \\ v_l p_{31} - p_{21} & v_l p_{32} - p_{22} & v_l p_{33} - p_{23} \end{bmatrix}} & \boxed{\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}} & = \boxed{\begin{bmatrix} m_{14} - u_r m_{34} \\ m_{24} - v_r m_{34} \\ p_{14} - u_l p_{34} \\ p_{24} - v_l p_{34} \end{bmatrix}} \\
 A_{4 \times 3} & \mathbf{x}_r & \mathbf{b}_{4 \times 1} \\
 \text{(Known)} & \text{(Unknown)} & \text{(Known)}
 \end{array}$$

Find least squares solution using pseudo-inverse:

$$A \mathbf{x}_r = \mathbf{b}$$

$$A^T A \mathbf{x}_r = A^T \mathbf{b}$$

$$\boxed{\mathbf{x}_r = (A^T A)^{-1} A^T \mathbf{b}}$$

3D Reconstruction with Internet Images

St. Peter's Basilica (1275 Images), Vatican City



[Snavely 2006]: Photo Tourism

3D Reconstruction with Internet Images

St. Peter's Basilica (1275 Images)



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